

# Day 13 Assignment — Ports, Protocols & Services Fundamentals

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## THEORY

### 1. What is a network port?

It is a logical communication endpoint and is used by services and applications, it is used to tell the device which service/ application should receive which traffic.

### 2. How many port numbers are available conceptually?

0-65535

### 3. What is a network protocol?

It is a group of rules to control network communications between devices.

### 4. What is TCP?

It is a transport layer protocol used to transfer packets reliably.

### 5. What is UDP?

It is a transport layer used to fastly transfer traffic but do not assure reliability

### 6. Explain two differences between TCP and UDP.

Tcp is connection-based while udp is connectionless

Tcp is slower but more reliable while udp is faster but less reliable

### 7. What is a listening port?

A listening port means a service is ready to communicate but not yet connected.

### 8. What is an ephemeral/client port?

### 9. What is a socket?

A socket is a communication endpoint that allows an application to send or receive data over a network

### 10. Which port is commonly used by SSH?

Tcp 22

### 11. Which port is commonly used by HTTP?

Tcp 80

### 12. Which port is commonly used by HTTPS?

Tcp 443

### 13. Which port is used by DNS?

Udp 53

### 14. Which UDP ports are commonly used by DHCP client/server?

67 and 68

### 15. What is TCP port 445 commonly associated with?

Smb (file and printing services)

**16. What is port 3389 commonly associated with?**

Remote desktop protocol (RDP)

**17. What does ss -tln display?**

It displays listening tcp and udp sockets status

**18. Does an open port automatically mean a vulnerability? Explain.**

No not necessary an open port means a service is available so its not automatically a vulnerability, so open ports are usually clues not evidence

**19. Why do firewalls use protocol and port information?**

Firewalls use them to define which network traffic is allowed or blocked

**20. Why are ports and protocols important in cybersecurity?**

They help security professionals identify services, monitor network traffic, and detect unauthorized or suspicious connections.

## PRACTICAL SCREENSHOTS

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ss -tuln  
Netid      State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port  
(kali@kali)-[~]  
$ ss -ltn  
State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port  
(kali@kali)-[~]  
$ curl -I https://example.com  
HTTP/2 200  
date: Wed, 23 Sep 2026 19:17:21 GMT  
content-type: text/html  
server: cloudflare  
last-modified: Tue, 22 Sep 2026 20:16:57 GMT  
allow: GET, HEAD  
accept-ranges: bytes  
age: 496  
cf-cache-status: HIT  
cf-ray: a3fbe39679b9e191-MRS  
(kali@kali)-[~]  
$
```